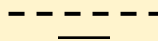


# Sumador De 6 Bits

Arquitectura de Computadoras



Actividad 2



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**Horario:** Mar y Jue de 9:00 a 11:00 A.M.

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**Fecha:** 24/09/2025

# Implementación de un sumador de 6 bits en MaxPlusII

```
suma6bits.vhd - Text Editor
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;
ENTITY suma6bits IS
    PORT ( Cin : IN STD_LOGIC;
          x, y : IN STD_LOGIC_VECTOR ( 5 DOWNTO 0 );
          s : OUT STD_LOGIC_VECTOR ( 5 DOWNTO 0 );
          Cout : OUT STD_LOGIC);
END suma6bits ;
ARCHITECTURE arq_suma OF suma6bits IS
    signal c : std_logic_VECTOR(6 DOWNTO 0);
    BEGIN
        PROCESS ( x, y, Cin )
        BEGIN
            c(0) <= Cin;
            FOR I IN 0 TO 5 LOOP
                s(I) <= c(I) XOR x(I) XOR y(I);
                c(I+1) <= (x(I) and y(I)) or (c(I) and (x(I) xor y(I)));
            END LOOP;
            Cout <= c(6);
        END PROCESS;
    END arq_suma;
END
```

Ilustración 1. Implementación de VHDL

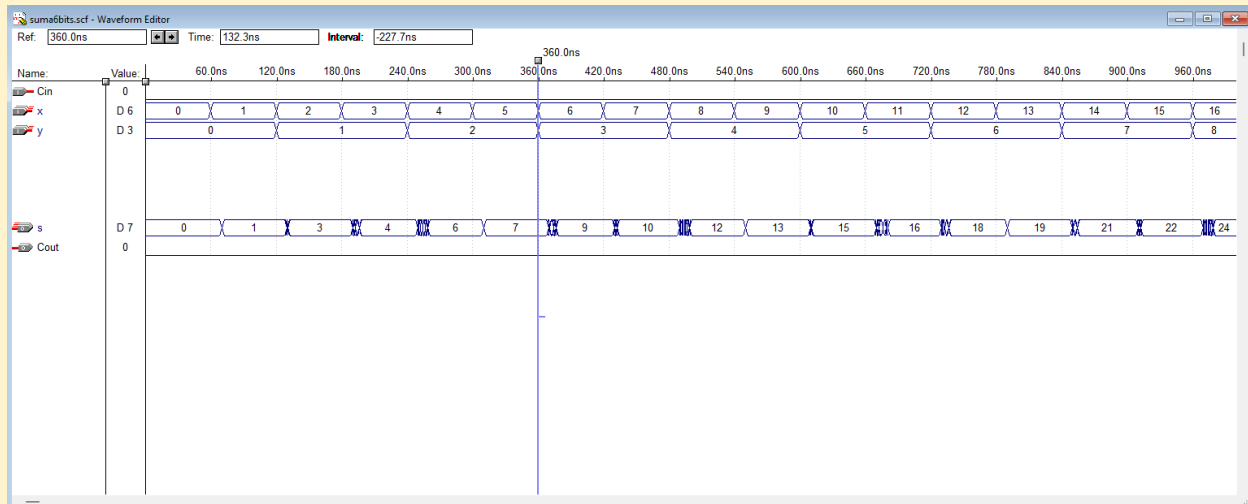


Ilustración 2. Simulación en MaxPlusII